PPARA Gene and Athletic Performance: A Study of Genetic Relationships through Sequence Analysis

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Introduction

Genetic factors play a critical role in shaping athletic performance. The PPARA gene is involved in regulating fat metabolism and energy use in muscle, which are essential for endurance and recovery (Eynon et al., 2013; Maciejewska-Skrendo et al., 2021).
Transcription factors such as TFE3, KLF6, and ATF3 influence skeletal muscle by controlling mitochondrial function, inflammation, and gene expression. These processes are important for physical adaptation (Wong et al., 2024; Fernández-Verdejo et al., 2017).
This research uses the bioinformatic tools VISTA (Frazer et al., 2004) and PROMO (Messeguer et al., 2002) to analyze conserved regions gene and predict transcription factor binding sites with potential roles in athletic traits.

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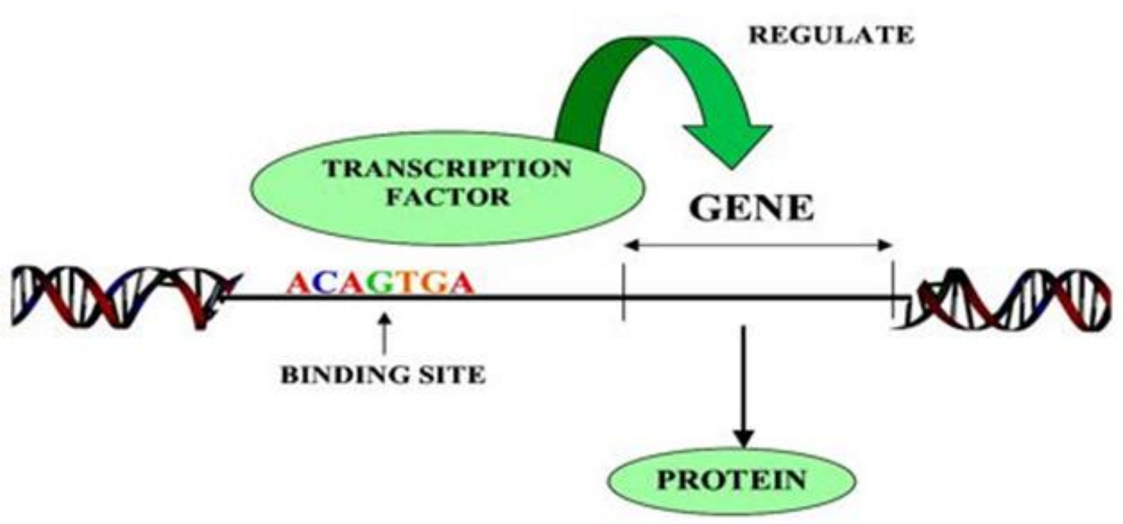

Figure 1. *Transcription factors bind to specific DNA sequences to regulate gene expression, a key mechanism explored in this study.*
Mubeen H. (2016). In silico approach to identify transcription factor binding sites and cis-regulatory elements in tubulin gene promoter.

Research Question

What transcription factors are predicted to bind to regulatory regions of the PPARA gene, and what potential relationships do they have with athletic traits such as speed, endurance, and strength?

Hypothesis

This research proposes that the PPARA gene contains predicted transcription factor binding sites associated with athletic performance traits. By identifying these predicted transcription factors, this study aims to better understand potential regulatory elements within the PPARA gene that may be related to traits such as endurance, speed, and strength.

References

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Baar K. (2004). PPARs in endurance adaptation. *Proc Nutr Soc*, 63(2), 269–273.
Draganidis D. et al. (2016). Inflammaging and skeletal muscle. *J Nutr*, 146(10), 1940–1952.
Eynon N. et al. (2013). ACTN3 and sprint performance. *Sports Med*, 43(9), 803–817.
Fernández-Verdejo R. et al. (2017). ATF3 in muscle adaptation. *FASEB J*, 31(2), 840–851.
Frazer K.A. et al. (2004). VISTA comparative genomics. *Nucleic Acids Res*, 32(Web), W273–W279.
Maciejewska-Skrendo A. et al. (2021). PPARA variant in gymnasts. *J Hum Kinet*, 79, 77–85.
Messeguer X. et al. (2002). PROMO binding site detection. *Bioinformatics*, 18(2), 333–334.

Methods

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Transcription Factor Identification

Predicted transcription factors potentially associated with athletic performance were identified through a literature review and bioinformatics database searches. Scientific literature was reviewed using the PubMed database to identify transcription factors previously reported in studies related to skeletal muscle function, metabolism, and exercise physiology. Additional analysis of potential transcription factor binding sites within the PPARA gene was conducted using the PROMO (ALGGEN) bioinformatics tool. Priority was given to transcription factors expressed in tissues relevant to PPARA activity, particularly skeletal muscle and metabolic tissues. Transcription factors identified through this process included ATF3, NF-κB, TFE3, KLF6, among others.

Comparative Genomic Analysis

Using VISTA Gateway, conserved noncoding sequences (CNS) were identified in the PPARA gene by comparing human ... regions are likely to play regulatory roles.

Obtaining Sequence Candidates

From the VISTA output, only CNS regions with ≥100 base pairs and ≥70% sequence identity were selected for further analysis. These thresholds ensured evolutionary conservation and potential regulatory significance.

Transcription Factor Binding Prediction

Selected CNS sequences were analyzed using PROMO ALGGEN to predict potential transcription factor binding sites based on sequence similarity and known motifs.

Results

Comparative genomic analysis identified multiple conserved noncoding sequences (CNS) within the *PPARA* gene. Transcription factors associated with metabolic regulation and skeletal muscle function were identified through literature review and bioinformatics analysis (Table I). VISTA comparative genomics identified CNS regions meeting the selection criteria of ≥100 bp and ≥70% sequence identity (Figure 2; Table II). PROMO analysis predicted transcription factor binding sites within these conserved regions, including ATF3 and NF-κB (Table III). An example of a predicted ATF3 binding site within a CNS region is shown in Figure 3.

Transcription Factor	Tissue Expression	Signaling Pathway	Function
TFE3	All fetal and adult tissues	Mitochondrial Adaptation	Supports mitochondrial function and endurance post-training
KLF6	Liver	Adipogenesis, interacting with PPARγ and C/EBPα	Regulates fat metabolism and storage; enhances adipocyte development
CREG1	Skeletal Muscle	Mitophagy	Maintains mitochondria and improves muscle endurance
ATF3	Skeletal Muscle	Inflammation Regulation	Reduces muscle inflammation; aids recovery and endurance adaptation
NF-kB	Skeletal Muscle	Inflammation Protein Synthesis	Promotes muscle repair by enhancing protein synthesis and reducing inflammation
NRF1, NRF2	Skeletal muscle	Mitochondrial Biogenesis, Fatty Acid Oxidation	Boost mitochondrial function and endurance; support metabolic health

Table I. Predicted transcription factors associated with the PPARA gene and their roles in skeletal muscle metabolism and athletic performance.

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Also, check figure captions to make sure the font style is consistent with the rest of the poster, Arial.

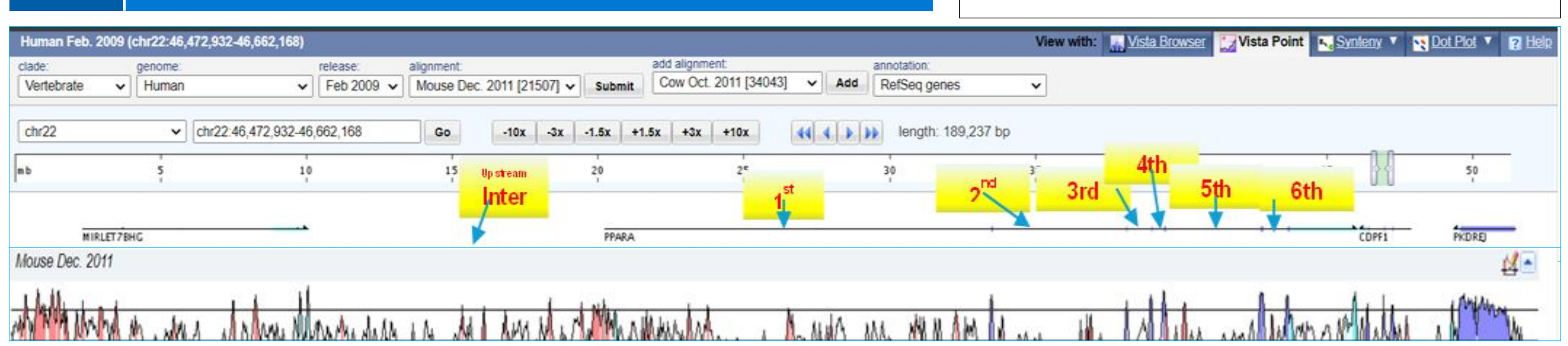

Figure 2. *VISTA comparative genomic analysis identifying conserved noncoding sequences in PPARA.*

Results

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Dissimilarity Rate:
13.49%

ATF3

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ATGAAGATACTTTGACATTTTCCTATGAGCATGGTGAAACAGGTTTAATTTGTATTAAAT
AGCTTGAAGCAATCCTTATTGGGAATTACAAGGTGGAATTTTAGTCACAGGAAAATAAA
GCATTTCACAAGCTACTTACTTTCATGAACAAACCAAACCTCTTCTTTACTGAGTCCTT
TAATTCTTCAGTGAATTCTCCAATTAAATAGGCCGAGACATTTTAGAAGTTTCCAGCAG
ACACCCACACTAGGCAGCTCCAGAGGCTTGTCCCAATTAGA
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Figure 3. *Predicted ATF3 binding site in CNS#6 (positions 100–107; 13.49% dissimilarity).*

Table II. Conserved noncoding regions of the PPARA gene identified using VISTA.

CNS	Gene	length	Identity	Location (Human Feb. 2009)
1	Intron 1	111bp	72.1%	chr22:46546588-46546698
2	Intron 1	213bp	73.2%	chr22:46556804-46557016
3	Intron 1	136bp	75.0%	chr22:46559170-46559305
4	Intron 1	144bp	72.2%	chr22:46589988-46590131
5	Intron 2	104bp	73.1%	chr22:46606236-46606339
6	Intron 5	282bp	76.2%	chr22:46618209-46618486
7	Upstream Intergenic	427bp	70.5%	chr22:46545078-46545530
8	Upstream Intergenic	137bp	78.8%	chr22:46546360-46546496
9	Upstream Intergenic	182bp	70.9%	chr22:46546093-46546297
10	Upstream Intergenic	259bp	72.6%	chr22:46545594-46545889
11	Upstream Intergenic	216bp	75.9%	chr22:46539579-46539831
12	Upstream Intergenic	113bp	77.0%	chr22:46534076-46534226
13	Upstream Intergenic	175bp	72.6%	chr22:46531435-46531649
14	Upstream Intergenic	163bp	72.4%	chr22:46528778-46528976
15	Upstream Intergenic	101bp	70.3%	chr22:46513861-46514001

Table III. Predicted transcription factor binding sites identified using PROMO.

CNS	DISSIMILARITY	BASE POSITION	TRANSCRIPTION FACTOR	LOCATION
1	10.03%	62-72	NF-kB1	chr22:46546588-46546698
2	11.69%	92-99	ATF3	chr22:46556804-46557016
3	11.69%	121-128	ATF3	chr22:46559170-46559305
5	11.69%	15-22	ATF3	chr22:46606236-46606339
6	13.49%	100-107	ATF3	chr22:46618209-46618486
10	6.58%	111-121	NF-kB1	chr22:46545594-46545889
11	13.49%	156-163	ATF3	chr22:46539579-46539831
13	10.84%	1-11	NF-kB1	chr22:46531435-46531649
15	9.80%	33-44	NF-kB	chr22:46513861-46514001

Discussion

Predicted binding of ATF3 and NF-kB within *PPARA* CNS regions suggests potential regulatory roles that may influence athletic performance. These factors are known to impact inflammation, recovery, and muscle adaptation. NF-kB has been linked to muscle maintenance by promoting protein synthesis and controlling inflammation (Draganidis et al., 2016). ATF3 supports endurance adaptations by reducing post-exercise cytokines (Fernández-Verdejo et al., 2017).
Their predicted presence in key regulatory regions of *PPARA* highlights a potential genetic mechanism behind performance traits. While these findings support the study's hypothesis, further experimental validation is needed to confirm their functional roles.